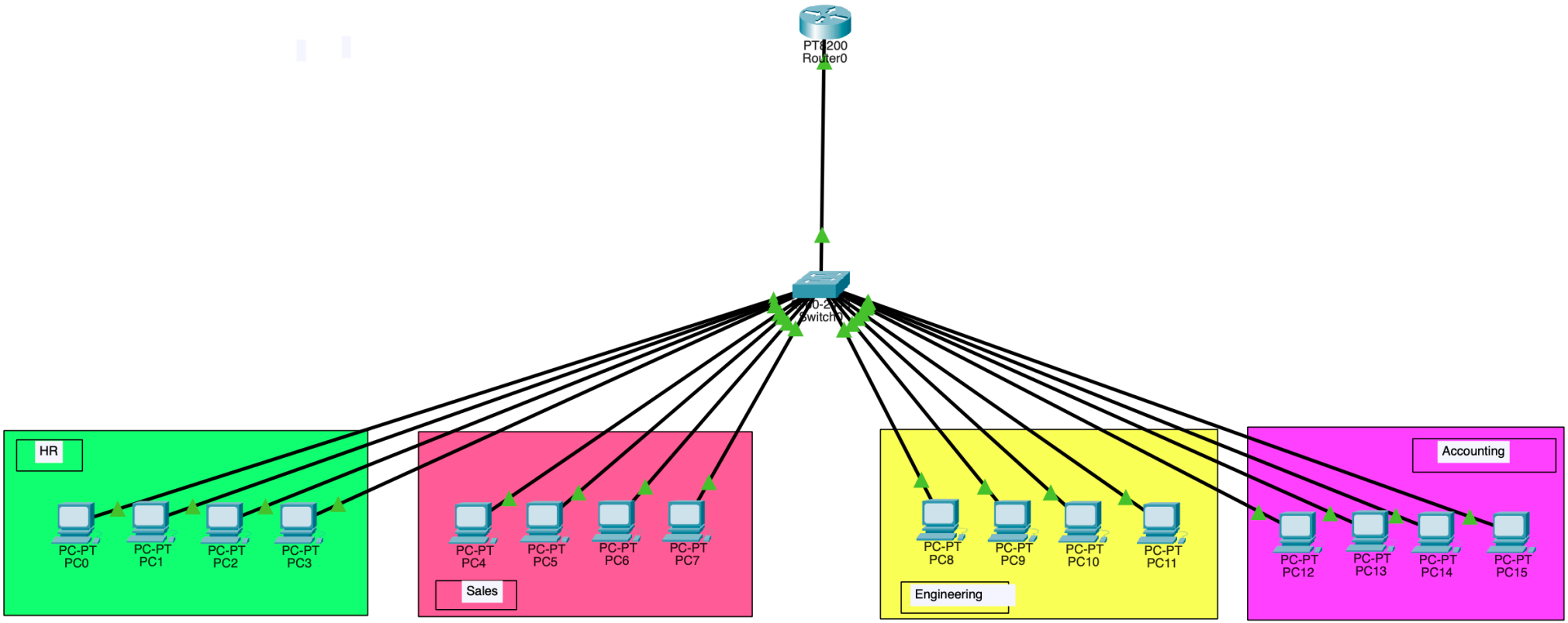


- Designed an enterprise-style network topology connecting 16 PCs across four departments: HR, Sales, Engineering, and Accounting.
- Created and configured separate VLANs to segment departmental traffic and isolate broadcast domains.
- Configured IP addressing and separate IP subnets for each departmental VLAN.
- Created and configured Switched Virtual Interfaces (SVIs) for each VLAN, assigning IP addresses to the virtual interfaces.
- Implemented Router-on-a-Stick inter-VLAN routing using router subinterfaces to enable communication between departmental VLANs.
- Configured IEEE 802.1Q encapsulation to identify and route traffic for multiple VLANs over a single physical router connection.
- Configured the switch-to-router connection as a trunk to carry traffic for multiple VLANs.
- Configured default gateways for departmental PCs to support communication across different IP subnets.
- Practiced VLAN creation, interface assignment, trunk configuration, subnetting, and Layer 3 routing concepts.
- Developed a scalable network design that separates departmental traffic while allowing inter-VLAN communication.